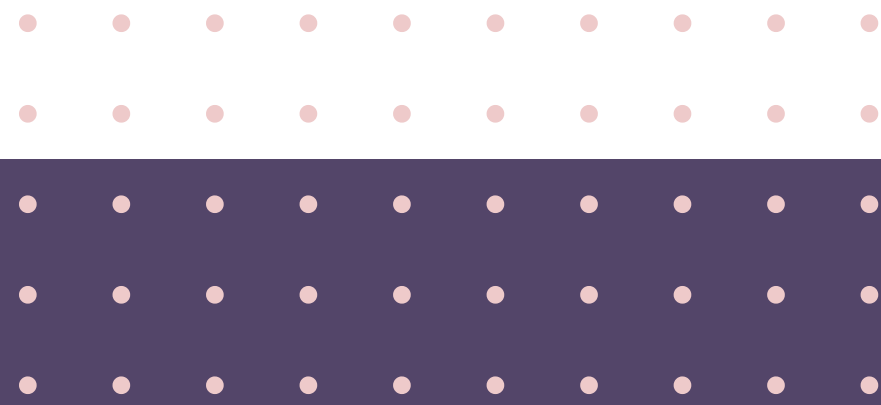




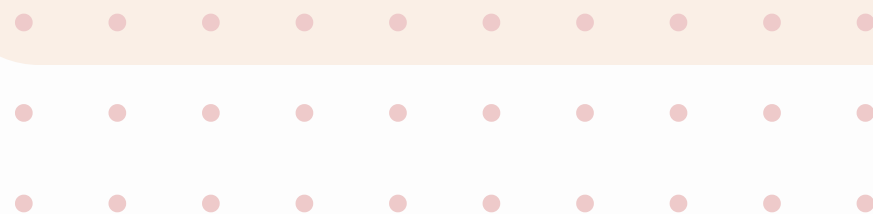
SMART GARDEN

Ben Smith, Jiana Kambo



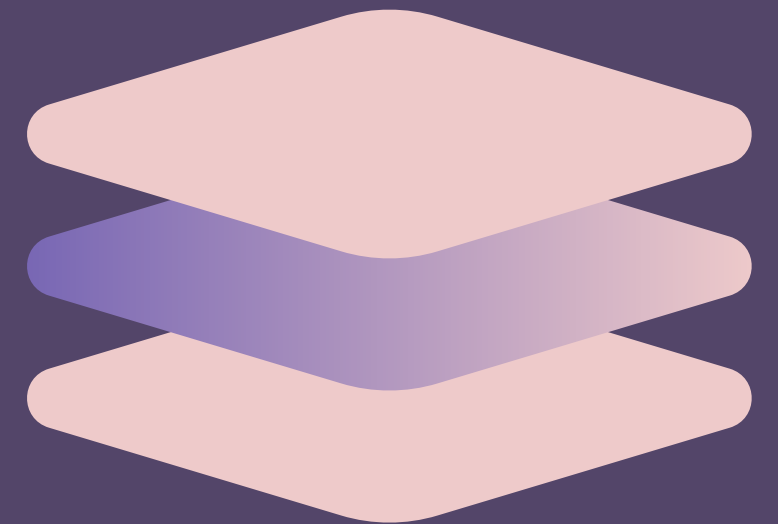
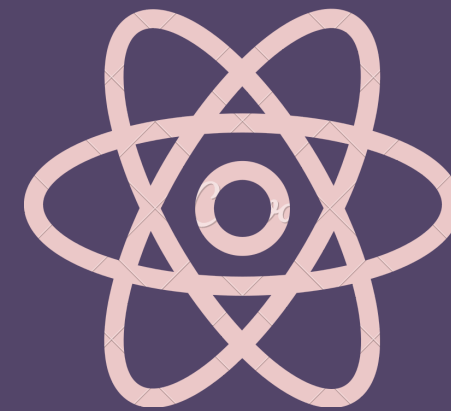
introduction

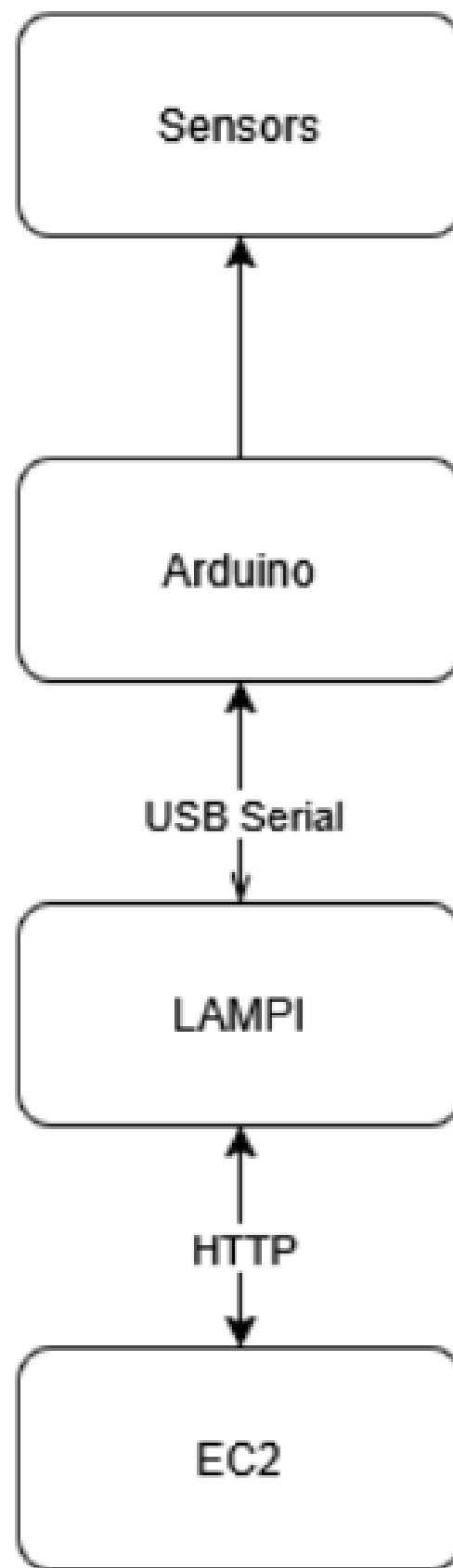
- Smart Garden is a connected device that:
- Monitors temperature, humidity, soil moisture, light, and water level
- Automates watering based on environmental conditions
- Uses Arduino for real-time processing
- Uses Raspberry Pi (LAMPI) as the front-end to interact with the system
- Uses Django website to display real time data, show historical trends and also control the water pump and set threshold values.



Setup

- Arduino collects data from:
- Soil Moisture Sensor
- Water Level Sensor
- Light Sensor
- DHT11 for Temperature and Humidity
- Pump connected through a Relay
- Use API endpoint to communicate bi directional from the django website and the pi
- Website shows information about the plant and lets you control the plant
- Real-world testing with a live plant

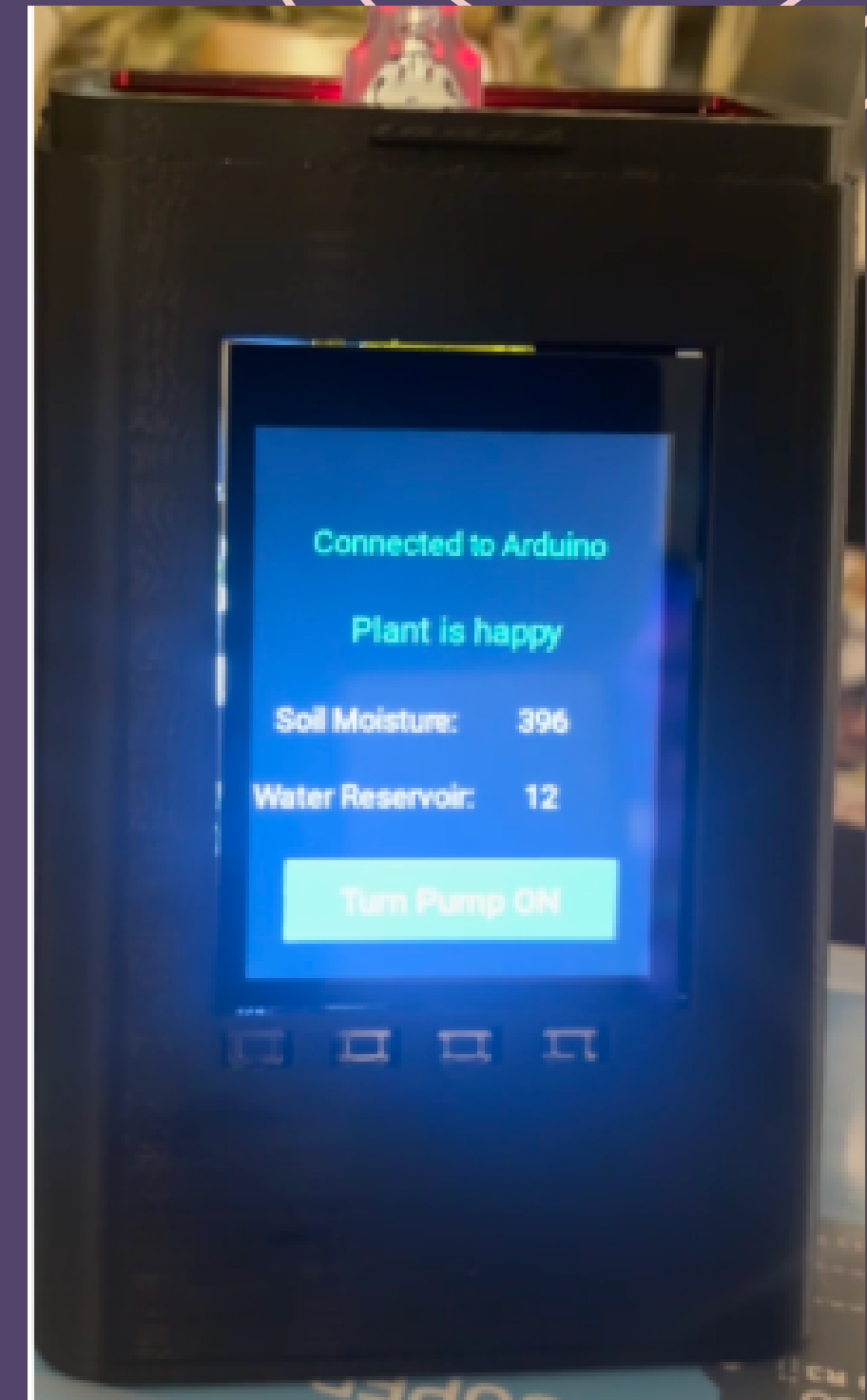




ARCHITECTURE

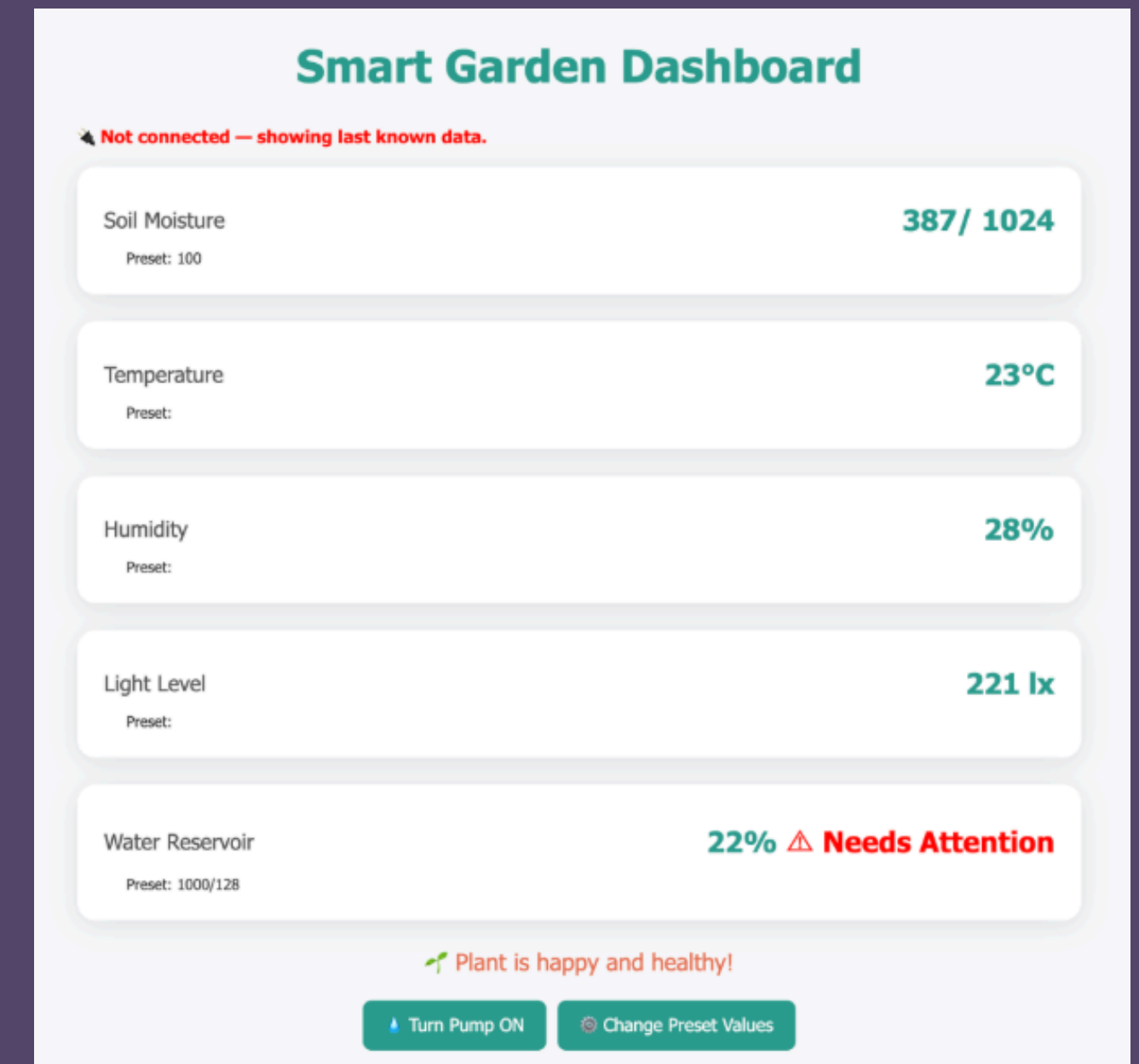
On the Pi

- Serial communication over USB
- JSON-formatted data exchanged every 2 seconds
- Kivy UI on Raspberry Pi displays:
 - Moisture, Reservoir Level
- User can toggle pump on/off via Pi
- Shows if you are connected to Arduino
- Only essential information shown here



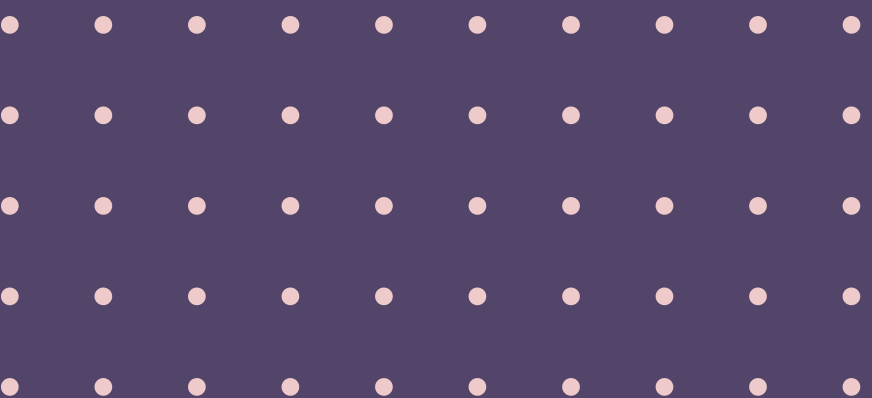
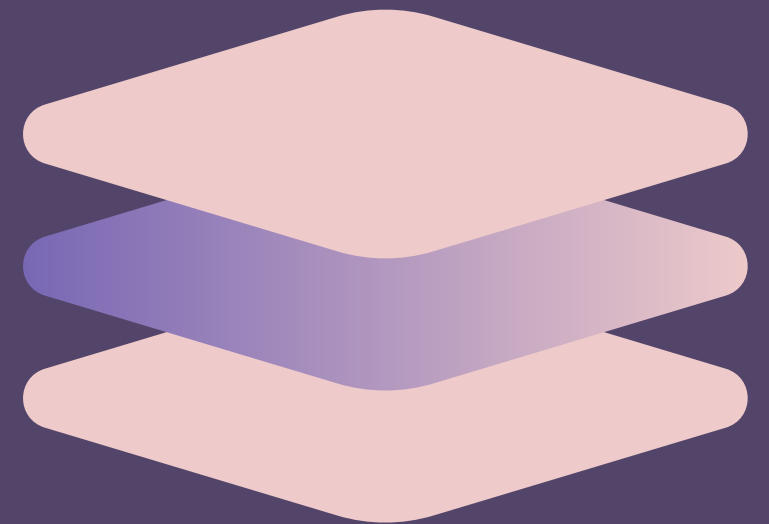
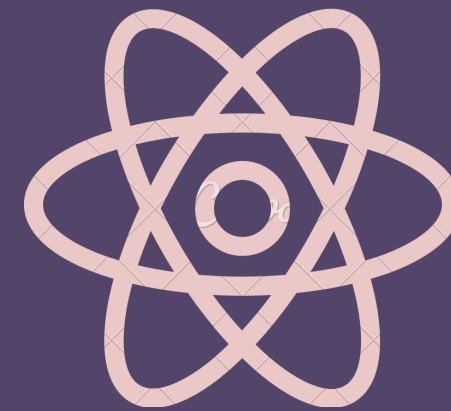
Website

- Shows real time data from the arduino unless not connected.
- Can water the plant manually from here as well
- Can change the preset values
- Shows what needs attention
- Preset value used for smart watering
- Shows graphical trends



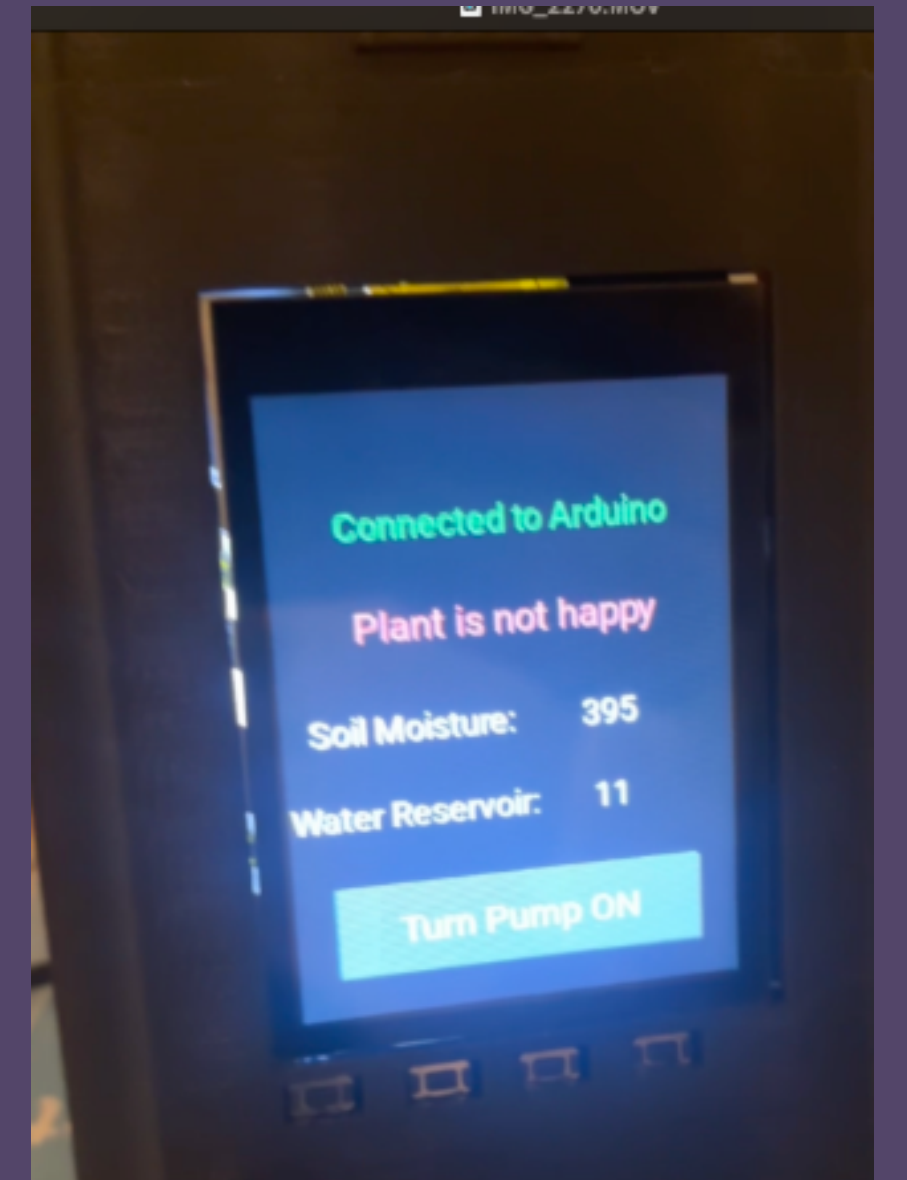
Smart Watering Logic

- Uses running average of last 10 readings
- Waits for moisture readings to stabilize before re-watering
- Prevents overwatering caused by premature readings



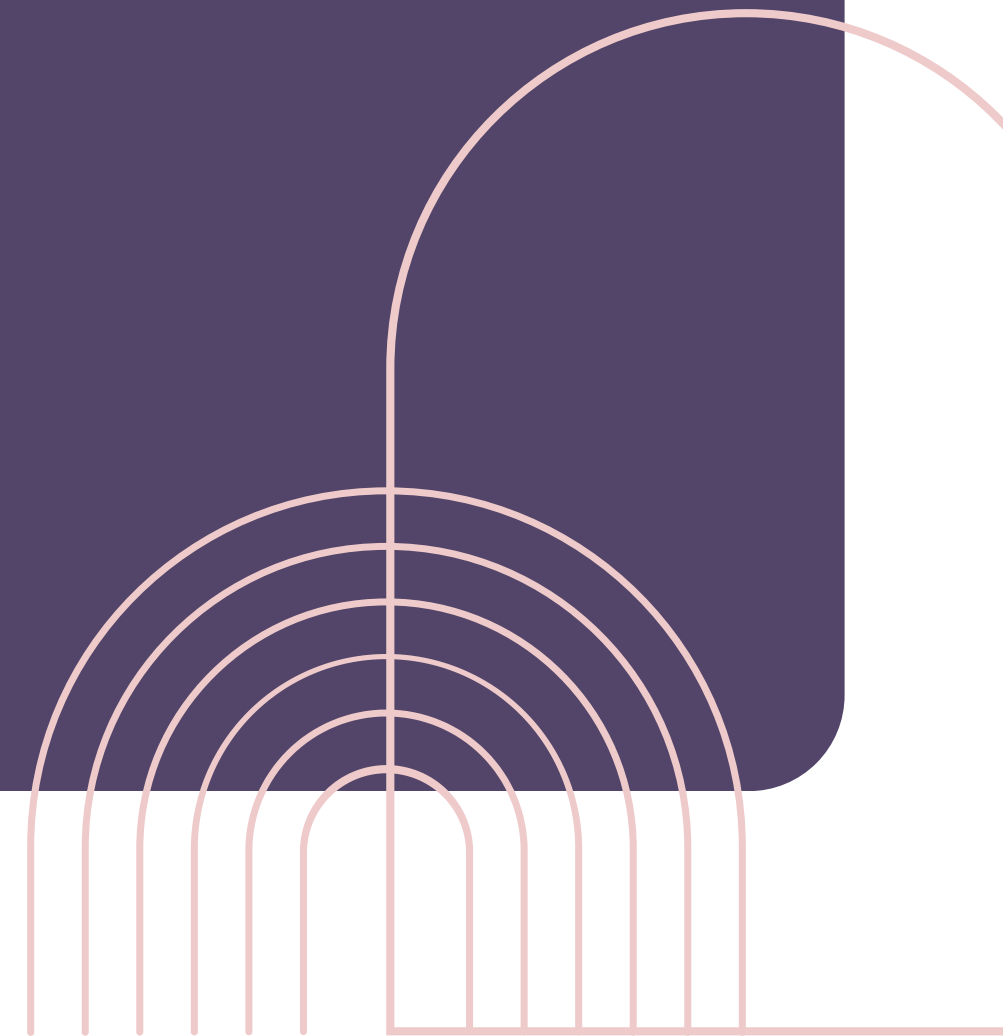
Plant Happy vs not

As long as the plant is above the soil moisture level we classify the plant is happy, if it is not that is communicated to the pi and the website and the user knows to give their plant the needed love



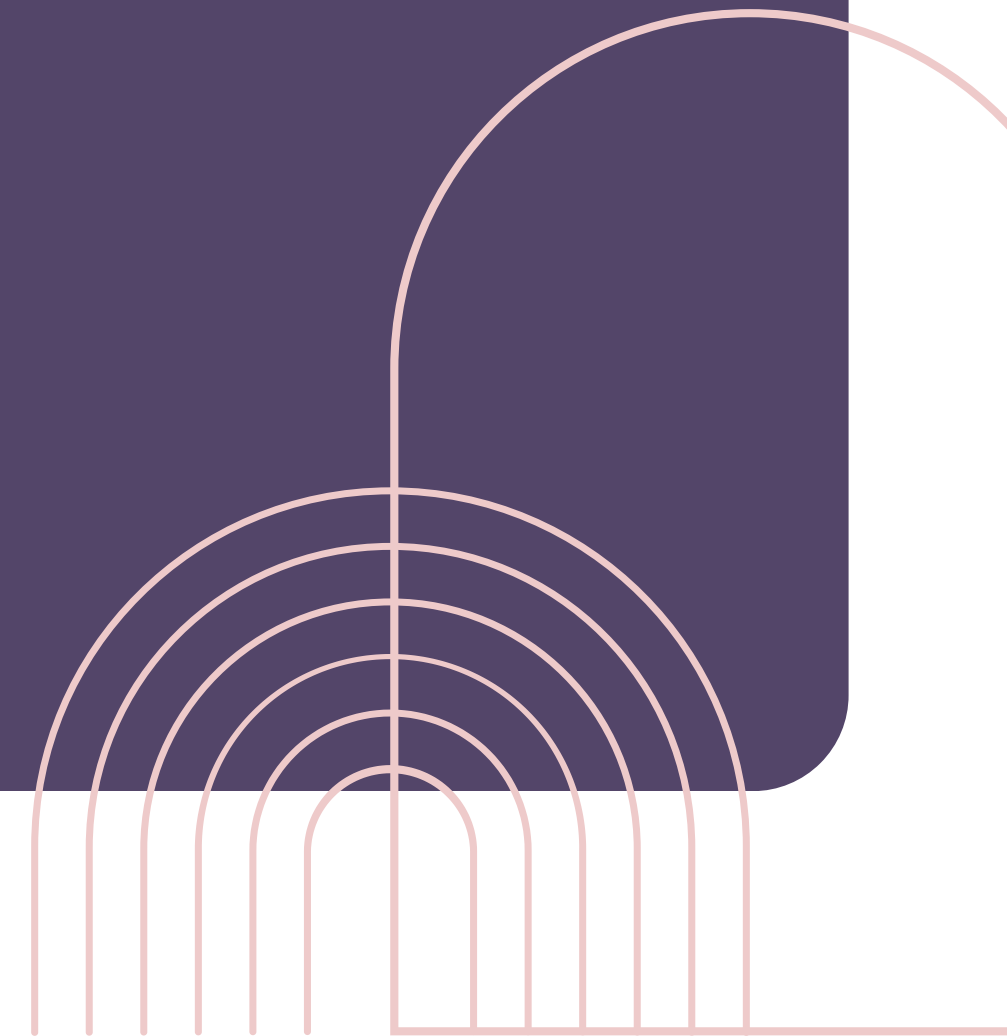
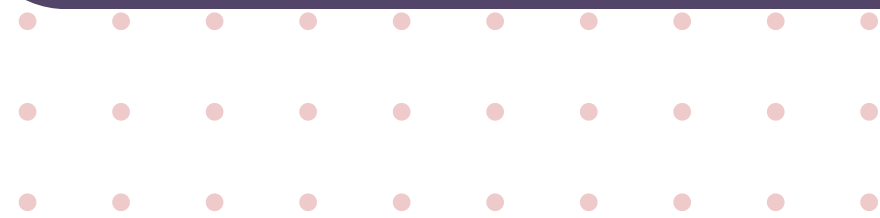
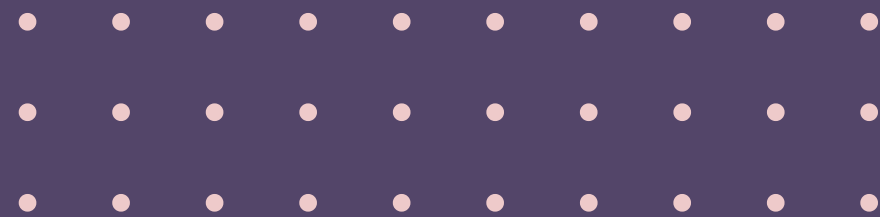
CHALLENGES WE FACED

- Getting the Pi to communicate with the arduino and the website through one port
- Took the demo video, everything worked, soldered it, it broke.



FUTURE WORK

- Building a nice enclosure for the device
- Supporting multiple plants



05.

Demo

